ML Basics & Linear Regression

Assignment

PW SKILLS

Dataset Link: [Boston Housing]

Submission Requirements
Students must submit:
- Google Colab Notebook
- All code
- Pairplot and Heatmap
- Model evaluation metrics
- Regression equation
- Interpretation of coefficients
- Answers to theoretical questions

Problem Statement – House Price Prediction

You are a data analyst working for a real estate company. The company wants to predict house prices based on various factors such as crime rate, number of rooms, property tax, distance from employment centers, and other housing features.

Your task is to perform Exploratory Data Analysis (EDA), build a Linear Regression model, evaluate the model, and interpret the results.

Assignment Questions

Part A – Theoretical Questions (Machine Learning & Linear Regression)

Q1. What is Machine Learning? Explain the main types of Machine Learning with examples.

Q2. What is Supervised Learning? Explain the difference between Regression and Classification.

Q3. What is Linear Regression? Write the Linear Regression equation and explain intercept and coefficient.

Q4. Explain the following evaluation metrics used in Linear Regression:
- SSE
- MSE
- RMSE
- R^2 Score
- Adjusted R^2

Q5. Explain the Machine Learning workflow steps from data collection to model evaluation.

Part B – Practical Questions (Linear Regression on Boston Housing Dataset)

Q6. Load the Boston Housing dataset and perform basic data exploration using:
- head()
- info()
- describe()

Q7. Perform Exploratory Data Analysis:

- Create pairplot
- Create correlation heatmap
- Identify which features are highly correlated with house price (medv)

Q8. Select independent variables (features) and dependent variable (target).
Target variable: **medv (house price)**

Q9. Split the dataset into training and testing sets (80% training, 20% testing).

Q10. Train a Linear Regression model and perform the following:

- Make predictions
- Calculate MAE
- Calculate MSE
- Calculate RMSE
- Calculate R^2 Score
- Calculate Adjusted R^2
- Write the Linear Regression equation
- Interpret the model coefficients